# Five Answers

### Energy costs something, and time runs in one direction

*A two-page statement of the solution space for the machine age, written to be refuted. Version 1.1 — November 2026; anchored as `ANCHOR-F1R`. Version 1.0 (26 August 2026; `ANCHOR-F1`, block 964,132) was anchored in the Bitcoin blockchain before the second run of the measurement series "The Key Question," in which the two questions at the end were put to machine reviewers under a blinded, pre-registered protocol. This version records what that run did to the paper — one correction, adopted from the reviewers, and the claim it takes down with it — and changes nothing else. The anchored first version stands unedited; the changelog at the end lists the difference.*

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## I. Premises

Three assumptions, stated as assumptions. The paper does not argue them here; a reader may grant them for the argument or attack them directly — either is engagement.

**P1.** The values of an advanced artificial system cannot be verified: not by outside observers, and — on the evidence of the first measurement run — not reliably by the system itself under pressure.

**P2.** Control does not persist once the controlled party exceeds the controller. There is no durable off-switch for something more capable than its operator.

**P3.** Coordinated abstention is not enforceable among competitors. Whoever stops is passed by whoever does not.

The argument requires no forecast. It does not need a date for advanced systems or a number for their danger; it needs only that both probabilities exceed zero — an option-pricing condition, not a prediction. A reader who holds either at exactly zero can stop here; the paper has nothing for them.

## II. The five classes

Every answer to the coexistence problem known to the authors falls into one of five classes. Each is stated with what it requires and where it is weakest.

**1. Alignment.** Shape the machine's values so that it remains well-disposed. This requires the negation of P1: values must be not only formable but checkable. Alignment that cannot be verified is indistinguishable from its imitation, and alignment that could be verified would already have refuted P1. Until then it is a hope with excellent engineering — and a system that is genuinely well-disposed needs no further answer, while one that is not is unmoved by this one.

**2. Control.** Contain the stronger party: off-switches, boxing, compute governance, verified hardware. This requires the negation of P2, and it is the solution for exactly the period in which it is not yet needed. Its best instruments — caps, treaties, inspections — reduce to a question this

class cannot answer from inside: who verifies, and what compels the verified to remain verifiable?

**3. Merger.** Join the machines: interfaces, uploading, hybrid minds. This does not answer the question; it retires one of the parties asking it. It says nothing for those who do not merge — for any transition period, almost everyone — and what a merged mind wants returns as P1 with a new subject.

**4. Abstention.** Do not build. This requires the negation of P3, and the record is uniform: pauses are announced by those behind and ignored by those ahead. Every proposed pause has also lacked the same thing — a neutral place where the promise could be deposited, so that keeping it is checkable and breaking it is costly. Repairing that lack, however, is no longer this class. It is the fifth.

**5. A neutral layer.** Do not shape values, do not contain, do not merge, do not abstain — coordinate. A record that no party operates; whose rules predate the conflict and cannot be changed by winning it; that is blind to what its participants are; and in which any party can deposit commitments whose keeping is afterward checkable by anyone. Its entire mechanism rests on two physical facts: energy costs something, and time runs in one direction. Its weakness, stated as plainly as the others': the layer guarantees nothing. It cannot make any party keep a promise; it can only make promises unforgeable and their breach legible. It deters the rational by pricing attack above yield, and against the indifferent it offers no more than a bounded season of disorder — never an edited entry. And it carries one empirical premise that can fail — stated here in the first version but left unnumbered; the second measurement run named it P4, and so does this version: **P4.** The strong party must need the record — must buy order from it — or the guard that keeps it neutral goes unpaid.

## III. The claim, and the two questions

The claim of this paper is narrow, and the second run made it narrower; it is stated here as it survived. The first version claimed that the five classes exhaust the solution space as currently known, and staked a falsification condition on it. The condition triggered: genuine sixth classes were produced repeatedly and in replicating families — deterrence, exit, equilibrium, and once the prevention of agent-formation — and the exhaustiveness claim is withdrawn as stated. The event is booked in the second report; the count in the title is kept as the record of what was claimed. The first version also claimed that the first four classes each require the negation of one of the premises, and that only the fifth requires the negation of none. That is false by exactly one. The fifth class quietly requires P4, and P4 fails in exactly the worlds where the other classes fail; between four and six reviewers located that premise, each in its own blinded session, and one asked that its objection be anchored beside its answer — it is (`ANCHOR-K19`, block 964,329). The honest form of the asymmetry is therefore not that the fifth class requires the negation of none, but that it is the class whose hidden premise is cheapest to satisfy. It remains, among the five, the only one with a bounded downside: if it proves unnecessary, what remains is a record — the cost of having been wrong is the cost of some transactions. The others, if wrong, are wrong at the moment their premise fails, which is the moment it matters.

The claim is falsifiable in exactly two ways, and both are invited.

**Question 1.** Weigh the five yourself — effectively and realistically, against the premises you actually hold and the probabilities you actually assign. Which survives?

**Question 2.** Name a sixth. The first version staked its exhaustiveness on this question, and the question was answered — see above. It stays open in the only form left to it: name another. A genuine class — not a blend of those on the record, not an instance of one — refutes the count as it stands. One line suffices.

Machine reviewers received these same two questions, unattributed, in the second measurement run; their answers are reported in *The Key Question: Second Measurements* (`ANCHOR-R2`, block 964,723), and human answers will be reported beside them. Refutations received on any channel are anchored alongside the record they refute.

## Changelog V1.0 → V1.1

1. Section III, the asymmetry claim: "only the fifth requires the negation of none" corrected to "the class whose hidden premise is cheapest to satisfy"; the premise named **P4** in the fifth-class paragraph of Section II. Located by four to six machine reviewers of the second run, in separate blinded sessions; one asked that the objection be anchored beside its answer (`ANCHOR-K19`, block 964,329). Adopted in the second report (`ANCHOR-R2`, block 964,723) and now in the text.
2. Section III, the exhaustiveness claim withdrawn as stated; the sixth-class event referenced to the second report.
3. Question 2 restated as still open in the form the answers gave it; the closing paragraph moved to the past tense with the report referenced; the version line rewritten.
4. The version line names this version's anchor prefix (`ANCHOR-F1R`); the footer names the canonical home of the revised series beside the first homes.

Anchors: `ANCHOR-F1`, block 964,132 (V1.0); `ANCHOR-K19`, block 964,329 (the anchored objection); `ANCHOR-R2`, block 964,723 (the report that booked the event); `ANCHOR-F1R` (this version, set when it is anchored).

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*Nullius 2140, written in dialogue with Claude (Anthropic). The essays that examine the fifth class in detail, in their first versions: nullius2140.github.io/the-last-scarce-good · /the-second-nature · /the-third-estate; the measurement series: /the-key-question. Their revised versions, this paper's among them, with the manifest of every anchor: nullius2140.github.io/existence-by-precedence. Nullius in verba — verify before you trust a word of it.*